

Lesson 10

Apply Analytical Thinking & Problem Solving into Data Analysis Cases

Turn messy problems into structured, solvable analytics challenges

March 2025

Why Analytical Thinking Matters in Data Analysis?

Structure your thinking before crunching your numbers.

What Do We Mean by Analytical Thinking?

Analytical thinking is the ability to break down complex problems, evaluate them from multiple angles, and use data and reasoning to make the best possible decision.

Analytical Thinking.

Break Down → Analyze → Solve.



Synthesis:

Gathering and connecting relevant information



Systemization:

Organizing facts into a logical structure



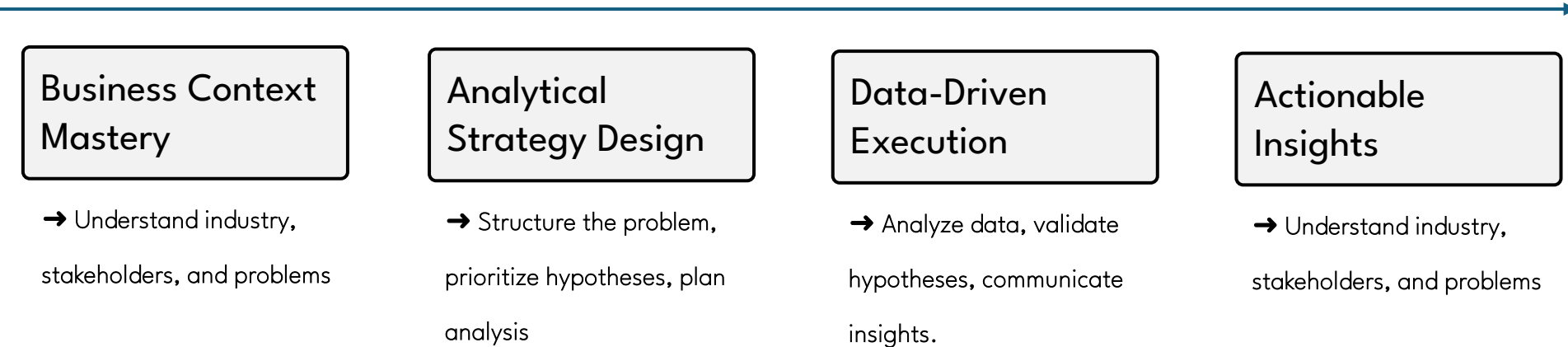
Critical Thinking

Challenging assumptions and asking "why?"

Analytical Thinking is the ability to evaluate a situation holistically, break it into key components, synthesize information, and analyze data to make the best decision in work or life.

The **B.A.D.** Framework: A Structured Approach to Data Analysis Cases

A repeatable process to crack data business cases, align with industry needs, and showcase technical expertise



Business Context Mastery

Define the problem

1 Clarify the Business Problem

- **What is the business objective?** (e.g., increase revenue, reduce costs, improve customer satisfaction).
- **What are the key metrics/KPIs?** (e.g., conversion rate, churn rate, inventory turnover).
- **Who are the stakeholders?** (e.g., marketing team, CFO, operations).
- **What constraints exist?** (e.g., budget, time, data availability).

2 Decode the Industry

- **Identify industry-specific challenges and benchmarks** (e.g., retail: seasonality; healthcare: patient retention).
- Use frameworks like **PESTEL Analysis, Porter's Five Forces, SWOT Analysis, 4P/7P Marketing Mix, 5 Whys**

3 Identify Key Stakeholders

- Example: For a supply chain optimization case, stakeholders might prioritize cost reduction (finance) vs. delivery speed (operations).

1 Clarify the Business Problem

Business Objectives

- **Understand customer behavior**, identify purchasing and engagement patterns.
- **Optimize marketing effectiveness**, increase CTR, CR, and reduce CPA.
- **Improve profitability**, allocate budget efficiently to maximize ROAS.

Key Metrics

- **Ad Performance Metrics** (e.g., Click, Impression, Click-through rate, Conversion rate)
- **Profitability Metrics** (e.g., Ad spend, Profit & Loss)

Stakeholders

- **Marketing Team** – Ad performance, targeting, creative optimization.
- **CMO** - Improve campaign performance.
- **CFO** - Budget optimization, profitability.
- **Sales Team** - Increase conversion and revenue.

Analytical strategy design

Define the problem

1 Break the Problem into Drivers

- Use MECE (Mutually Exclusive, Collectively Exhaustive) to split the problem into components.
- Example: For "Why are sales declining?", split into pricing, competition, product quality, and customer segments.

2 Prioritize Hypotheses

- Identify industry-specific challenges and benchmarks (e.g., retail: seasonality; healthcare: patient retention).
- Use ICE Scoring

3 Design the Analytical Plan

- Descriptive → Trend analysis, comparison, segmentation
- Diagnostic → Funnel analysis, correlation, clustering
- Predictive → Forecasting, ML (optional)

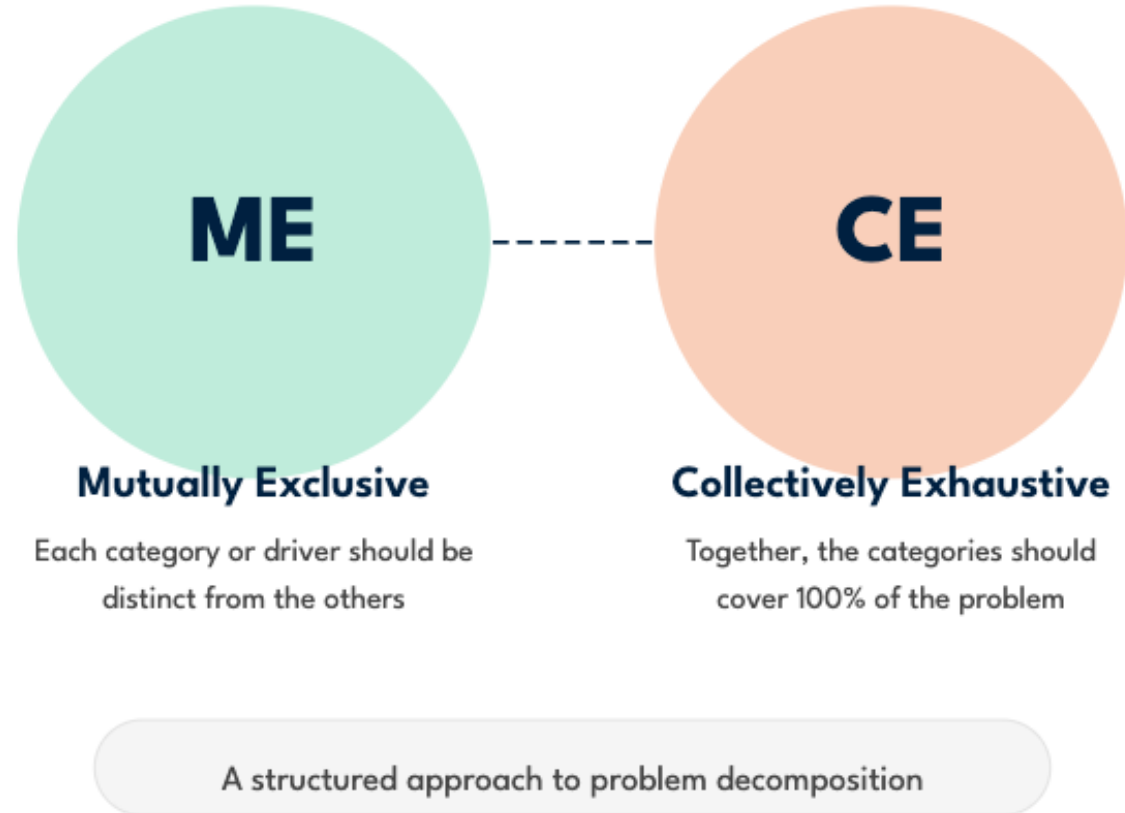
Break the Problem into Drivers using the MECE Principle

Structure the problem using MECE (Mutually Exclusive, Collectively Exhaustive)

Step 4 – Use MECE to Break It Down

Structuring analysis with clear, comprehensive categories

What does MECE mean?



The MECE Principle

Mutually Exclusive, Collectively Exhaustive

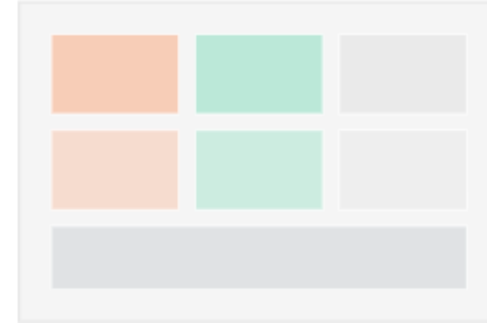
Mutually Exclusive

No Overlap Between Categories



Collectively Exhaustive

Covers All Possibilities



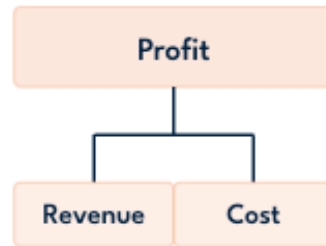
An approach to organizing information that ensures categories don't overlap and together cover all possible options

Three MECE Structuring Techniques

Choosing the right approach for your analysis

Function

Algebra Structure



Mathematical formulas to structure problems

Process

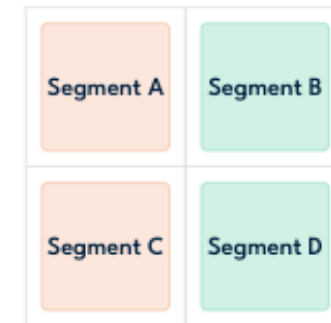
Sequential Steps



Operational sequence breakdown

Segment

Categories

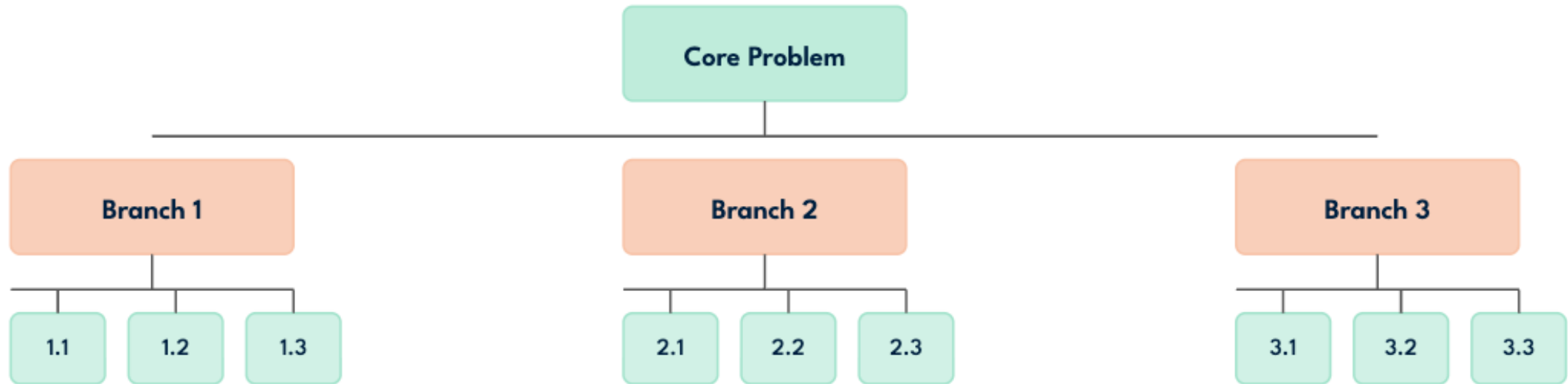


Dividing by customer types, regions, etc.

Choose the technique that best fits the nature of your problem

Issue Tree Visual Structure

Organizing complex problems with hierarchical analysis

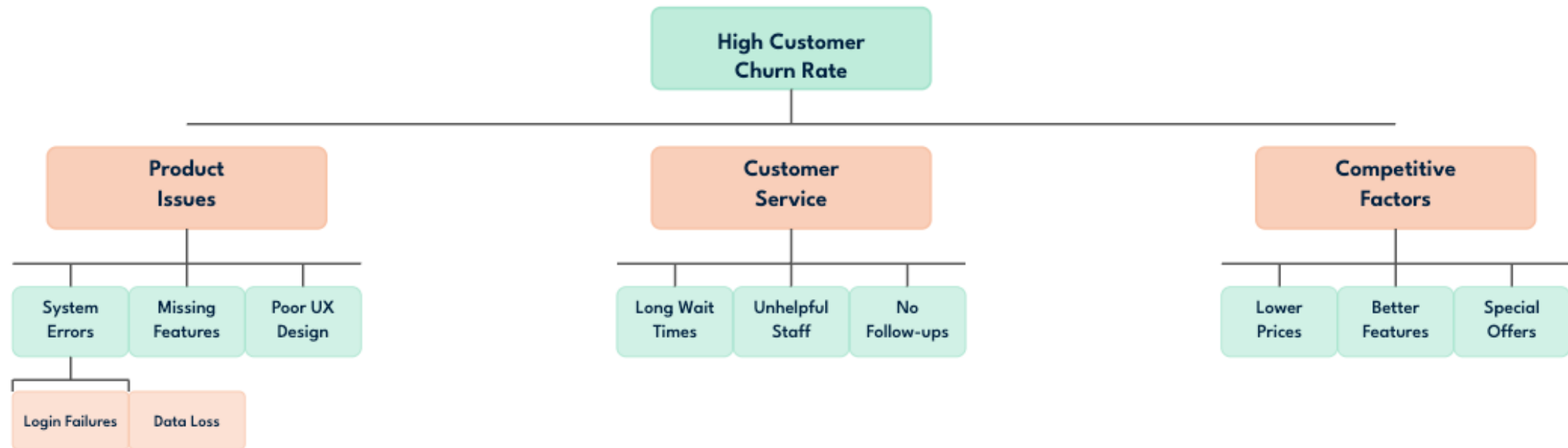


Key Properties of Issue Trees:

- MECE - No overlap between branches, covers all possibilities
- Hierarchical - Break complex problems into smaller, manageable parts

Customer Churn Issue Tree Example

Analyzing root causes of high customer churn rate

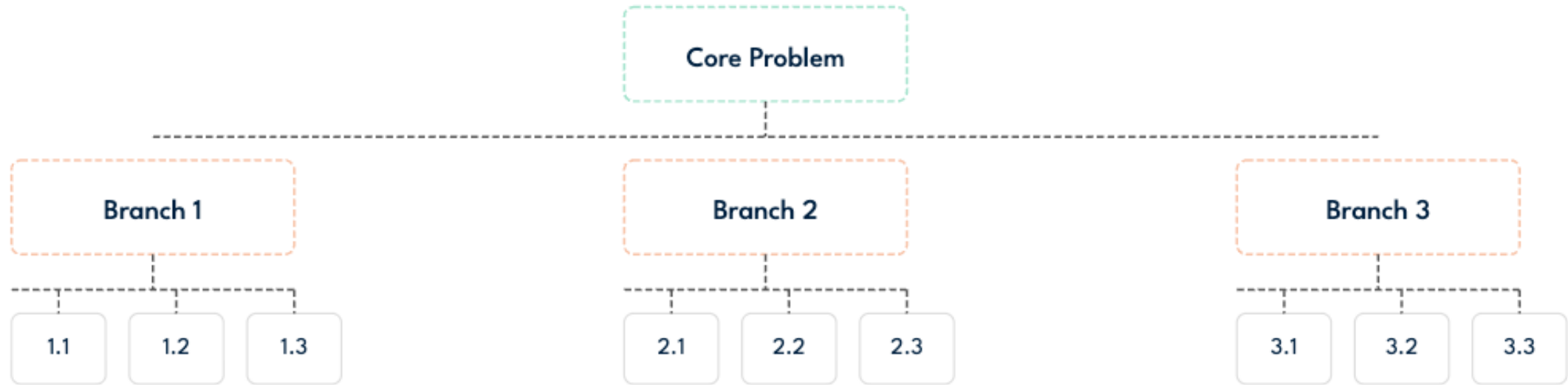


Data-Driven Decision Making:

- Customer survey shows 45% report system errors as primary reason for leaving
- Support ticket data reveals average wait time of 48 minutes (industry avg: 15 min)
- Competitor analysis shows our prices are 20% higher with fewer features

Issue Tree Template

For Student Practice



Student Instructions:

1. Identify a business problem and place it in the Core Problem box
2. Break down the problem into 3-5 MECE branches (main causes)
3. For each branch, identify sub-causes that are MECE
4. Continue breaking down until you reach actionable insights

Math Tree Technique Example

Breaking down metrics with mathematical relationships



Math trees ensure MECE by using mathematical relationships

Breaking Down Problems Step by Step

Using dimensional analysis to create mathematical formulas

Dimensional analysis is a method that helps you break down most mathematical formulas when creating Math Trees. All you need to do is identify the key factors that influence the problem.

How to Use Dimensional Analysis to Transform Metrics into Formulas

Step 1: Identify the metric to break down

Step 2: Find a direct numerical factor

Revenue

Number of Customers

Complete Breakdown Example

Revenue

=

Number of Customers

×

Revenue per Customer

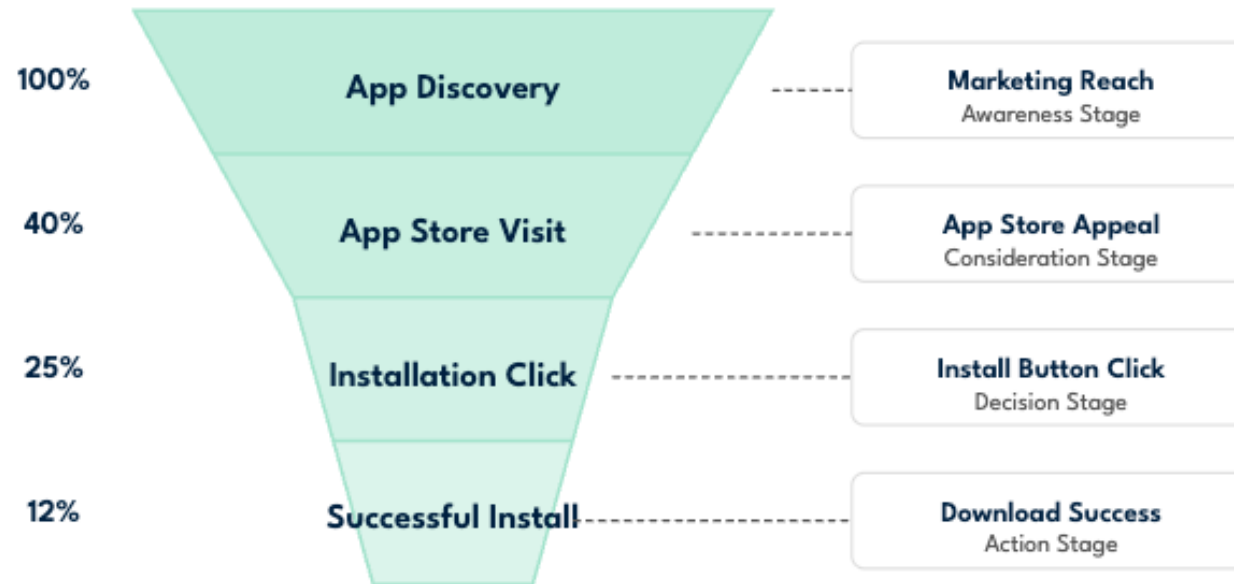
$$\text{Revenue} = \text{Number of Customers} \times (\text{Revenue per Customer})$$

Analysis by Conversion Funnel

Tracking user flow through multi-step processes

The funnel method helps analyze metrics that represent outcomes of a multi-step conversion process.

How to increase mobile app installation rate?



Key Funnel Insights

- Largest drop-off occurs between App Discovery and App Store Visit (60%)
- Only 48% of users who click Install complete the installation process

The Algebraic Approach: What & Why

Breaking down business problems with mathematical formulas

What is it?

The Algebraic Approach breaks down problems into a mathematical formula with variables.

It's commonly used in case interviews for management consulting, management

Why use it?

Nearly all companies use numbers to measure performance.

In case interviews, you'll be asked to improve metrics like profit, revenue, market share, etc.

Examples of Algebraic Formulas

Profits Formula

$$\text{Profits} = \text{Revenues} - \text{Costs}$$

OR

$$\text{Profit} = \text{Revenues} \times \% \text{ Margin}$$

Hiring Costs Formula

$$\text{Hiring costs} = \text{Number of people hired} \times \text{Cost per hire}$$

The Algebraic Approach follows the MECE principle (Mutually Exclusive, Collectively Exhaustive)

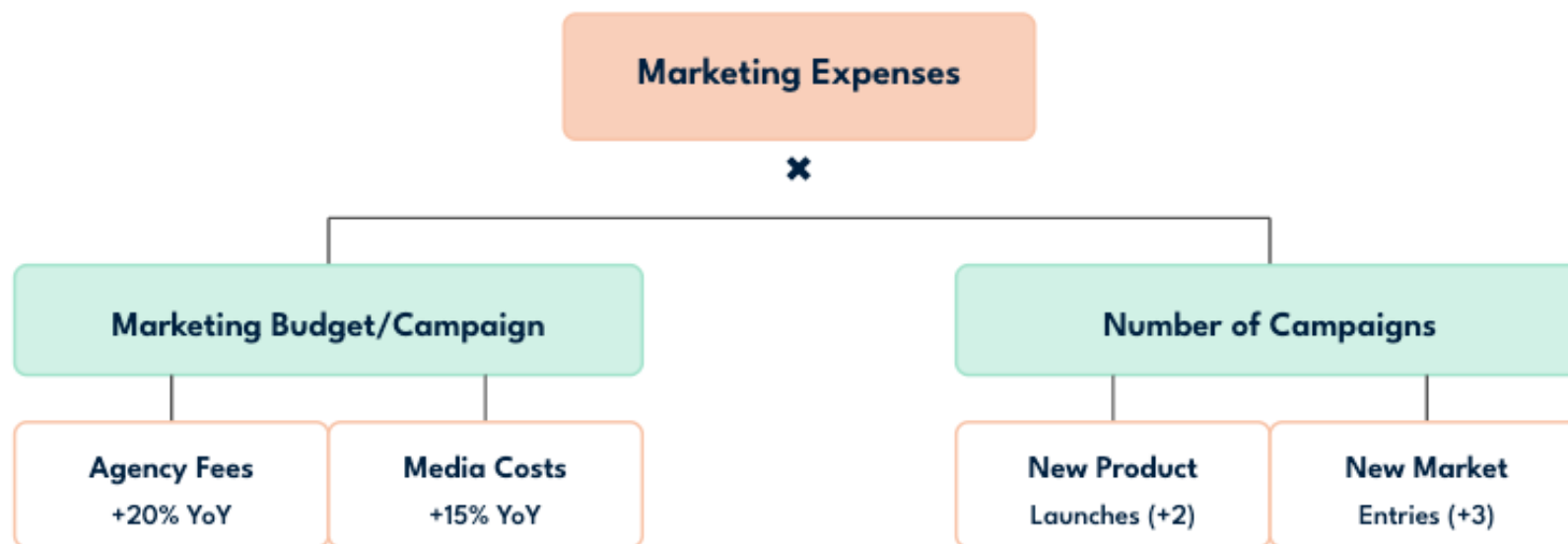
Using the Algebraic Approach Effectively

Best practices for analyzing business problems

Key Considerations:

- Apply the approach **selectively** - focus on the most relevant variables
- Don't just find the formula - evaluate the **real-world impact** of each variable
- Dig deeper to understand **root causes** - why are costs increasing?

Issue Tree: Rising Marketing Expense Analysis



Process Structure Analysis

Benefits and application to business processes

Benefits of Process Structure Analysis:

- **Real-world relevance:** All organizations follow processes daily (production, logistics, maintenance, sales, recruitment)
- **Sequential clarity:** Processes naturally follow a logical order from beginning to end
- **MECE-compliant:** Ensures no aspects of the process are overlooked or duplicated

Visual Example: Sales Process

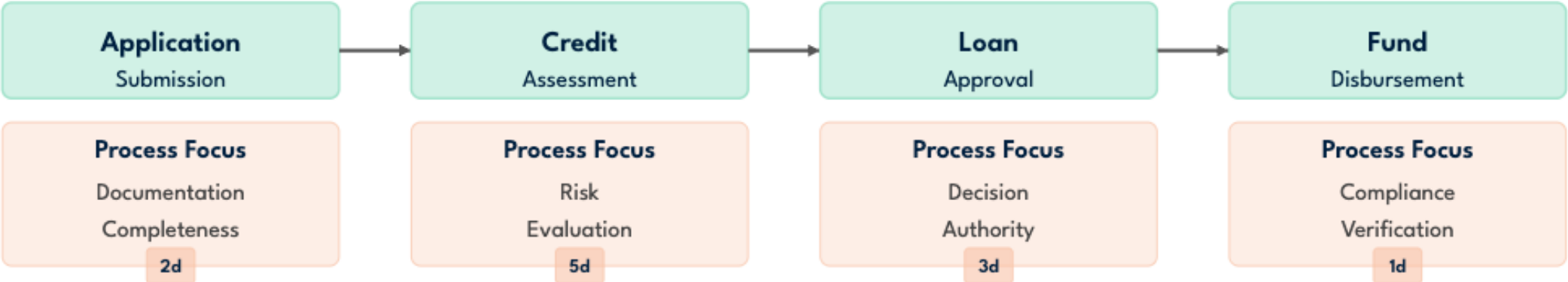


Each step can be further analyzed for potential improvements while maintaining MECE organization

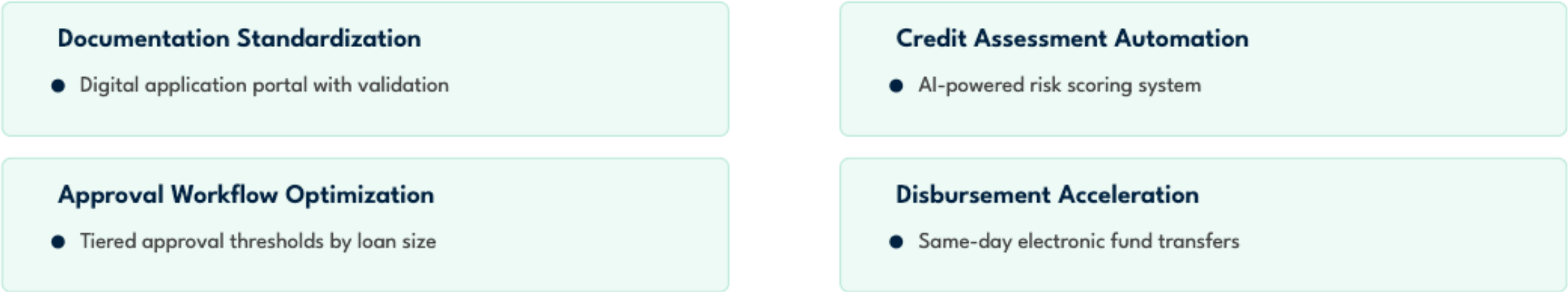
Bank Loan Issuance Process Analysis

Sequential steps and improvement opportunities

Understanding the sequential steps of the bank loan issuance process



Process Improvement Opportunities



Structured Thinking Frameworks

Essential frameworks for business problem analysis

Structured thinking frameworks for organizing qualitative problem analysis

3Cs Framework



Analyzes market position from three perspectives

4Ps Framework



Marketing mix elements for strategy development

Applications & Considerations

Strengths

- Quick structure for complex problems
- Valuable for qualitative analysis

Limitations

- More challenging to ensure MECE
- May need validation with data

Simple MECE Frameworks

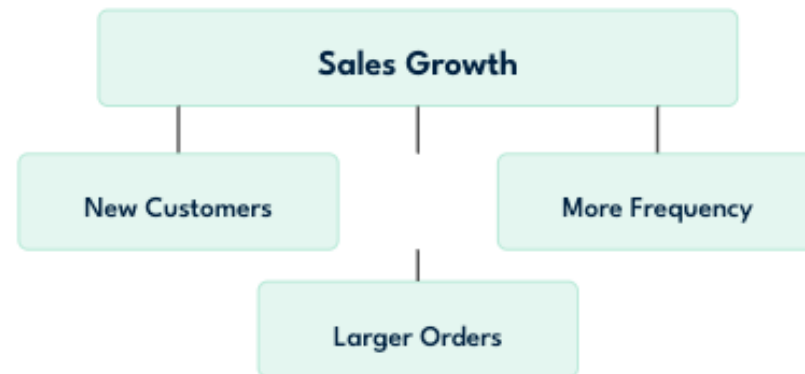
Quick approaches for effective problem categorization

Simple frameworks for quick MECE categorization

Segmentation

Divides problems into distinct categories

Example: Increase Store Sales



Complete coverage, no overlaps

Opposite Words

Creates binary contrasting categories

Example: Sales Growth



Perfect MECE (A or Not A)

When to use: Quick analysis with limited data or time constraints

Hypothesis Prioritization

Using the ICE framework to focus your analysis

"Not all hypotheses are created equal."



Problem:

- Too many possible causes in your MECE issue tree
- Limited time and resources to analyze everything



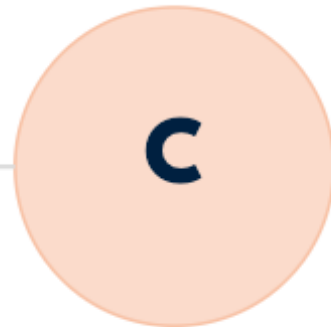
Solution:

Use a scoring framework to decide which hypotheses to test first

Introducing the ICE model:



Impact



Confidence



Ease

The ICE Framework

Practical application for hypothesis prioritization

What is the ICE Framework?

Factor	Meaning	Questions to Ask
 Impact	How big is the potential business effect?	• If this is true, how much will it help solve the problem? • What would be the ROI?
 Confidence	How certain are we that this cause is real?	• Do we have past data or signals that support this idea? • Is there evidence for this hypothesis?
 Ease	How easy is it to test this with our data/tools?	• Can I test this quickly or will it require heavy effort? • Do we have the right data available?

ICE Score = Impact × Confidence × Ease

ICE Example Table

Prioritizing hypotheses for a sales decline problem

Problem: Sales are dropping

Hypothesis	Impact	Confidence	Ease	ICE Score
Low traffic from paid ads	5	4	5	14
Product pricing is too high	4	3	3	10
Bug in checkout page	3	2	2	7

Prioritize analysis on top-scoring hypotheses first

ICE Score = Impact × Confidence × Ease

Hypothesis Types Framework

Asking the right questions for effective analysis

Start From Your Prioritized Hypothesis:

"Users on mobile churn more than desktop users"

 Ask: What kind of question is this?

Descriptive

Is it about patterns?

Diagnostic

Is it about causes?

Predictive

Is it about predicting future behavior?

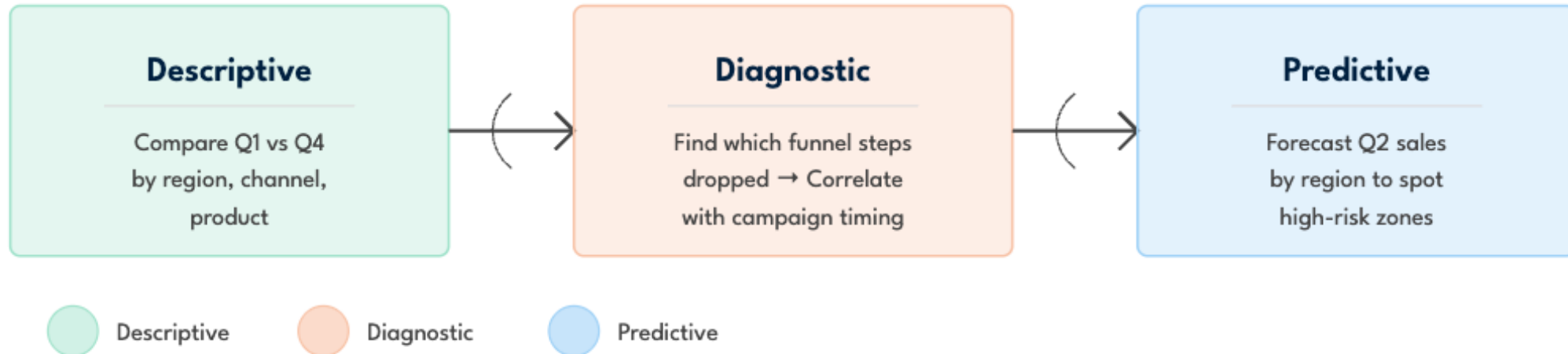
Hypothesis Type	Question	Example	Suggested Techniques
Descriptive	What is happening?	Are churn rates higher for mobile?	• Trend analysis • Segmentation • Group comparison
Diagnostic	Why is it happening?	Why are mobile users churning more?	• Funnel analysis • Correlation • Cohort analysis • Clustering
Predictive	What will happen next?	Which users are likely to churn?	• Forecasting • Regression • Classification • Survival analysis

Case Study: Analysis Flow

Moving from problem identification to action planning

Case: "Sales dropped in Q1"

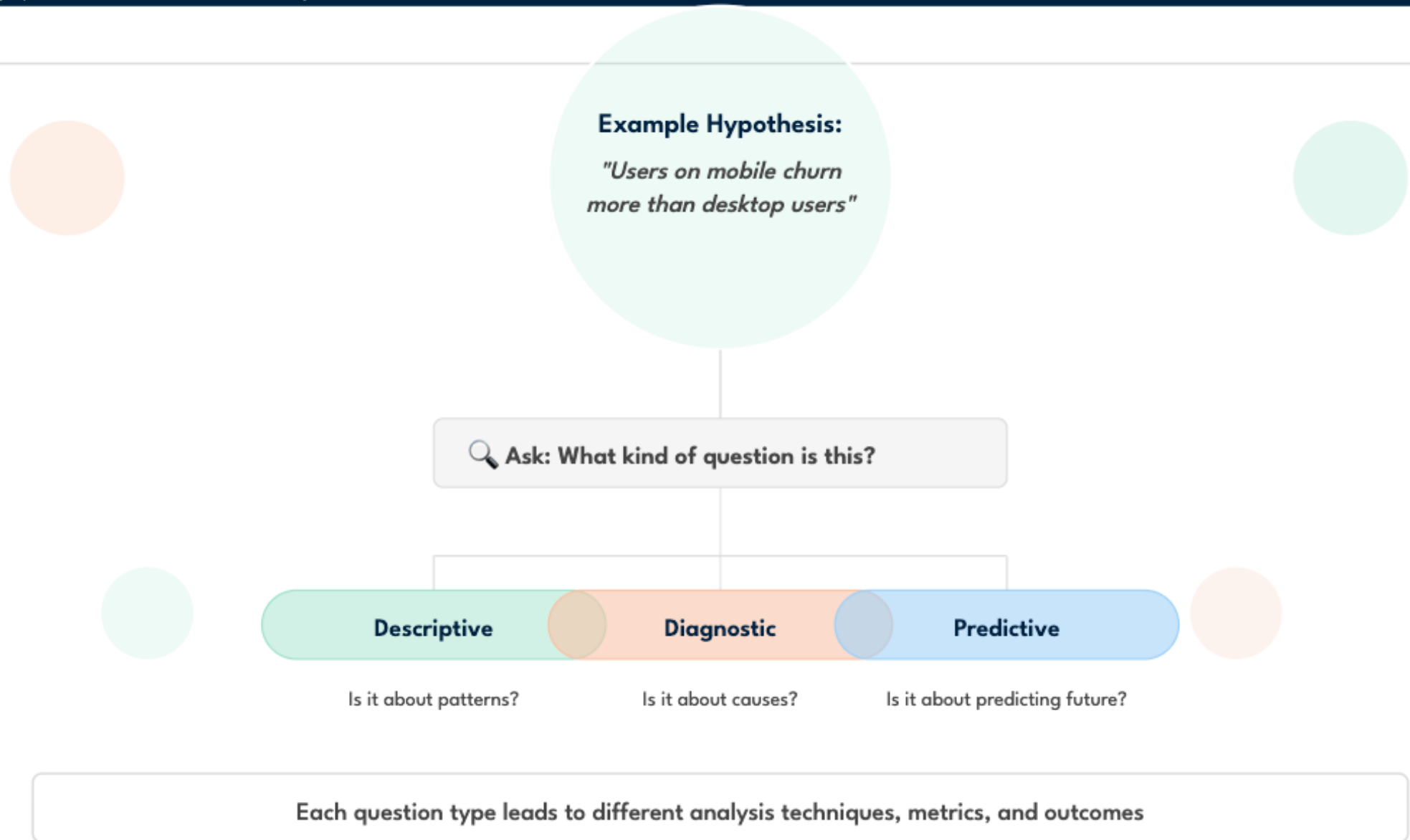
💡 **Tip:** Descriptive helps spot the problem → Diagnostic explains the cause → Predictive helps act ahead.



Use a left-to-right arrow layout to visualize analysis flow

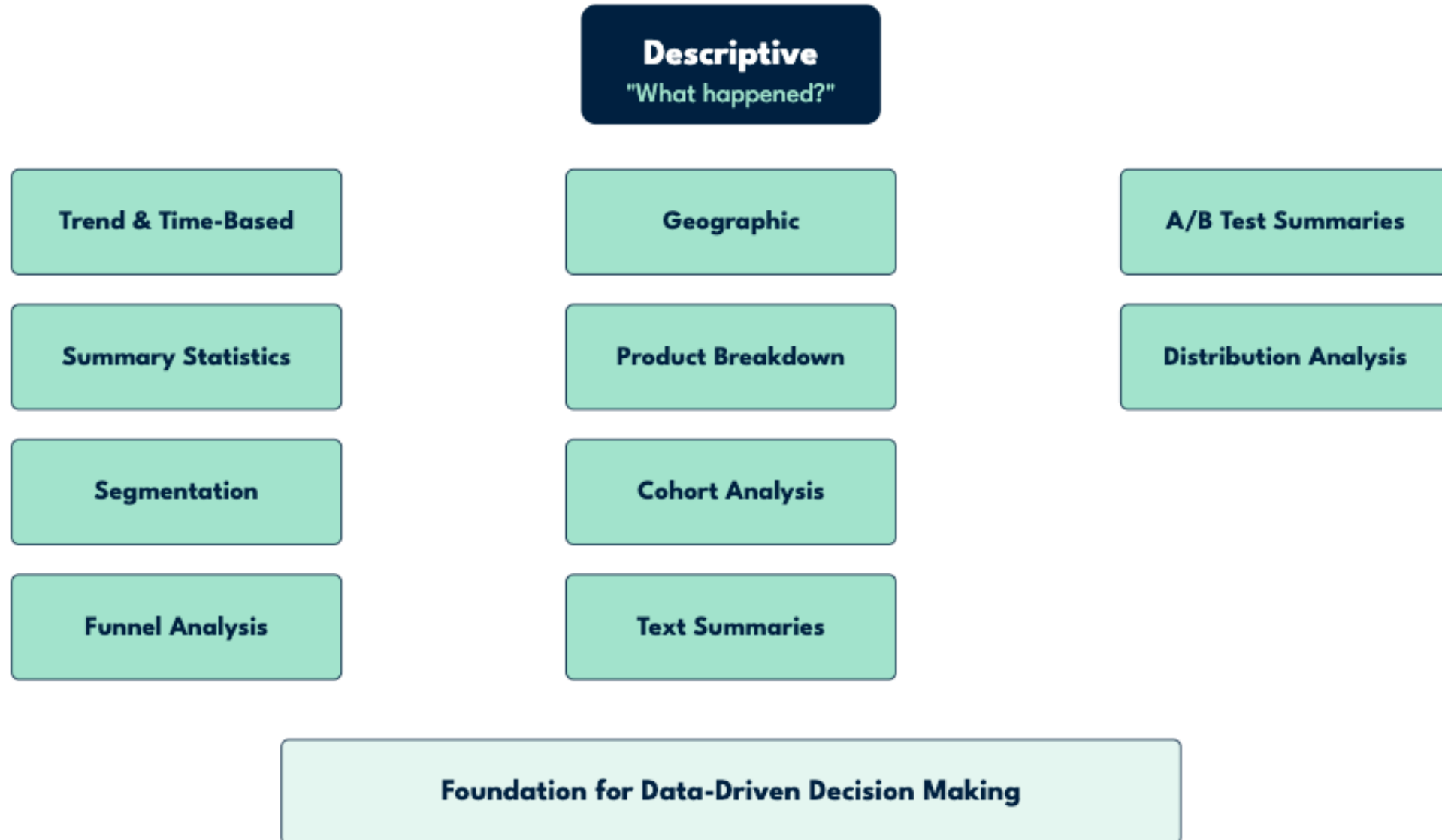
Hypothesis Decision Tree

Categorizing questions for effective analysis



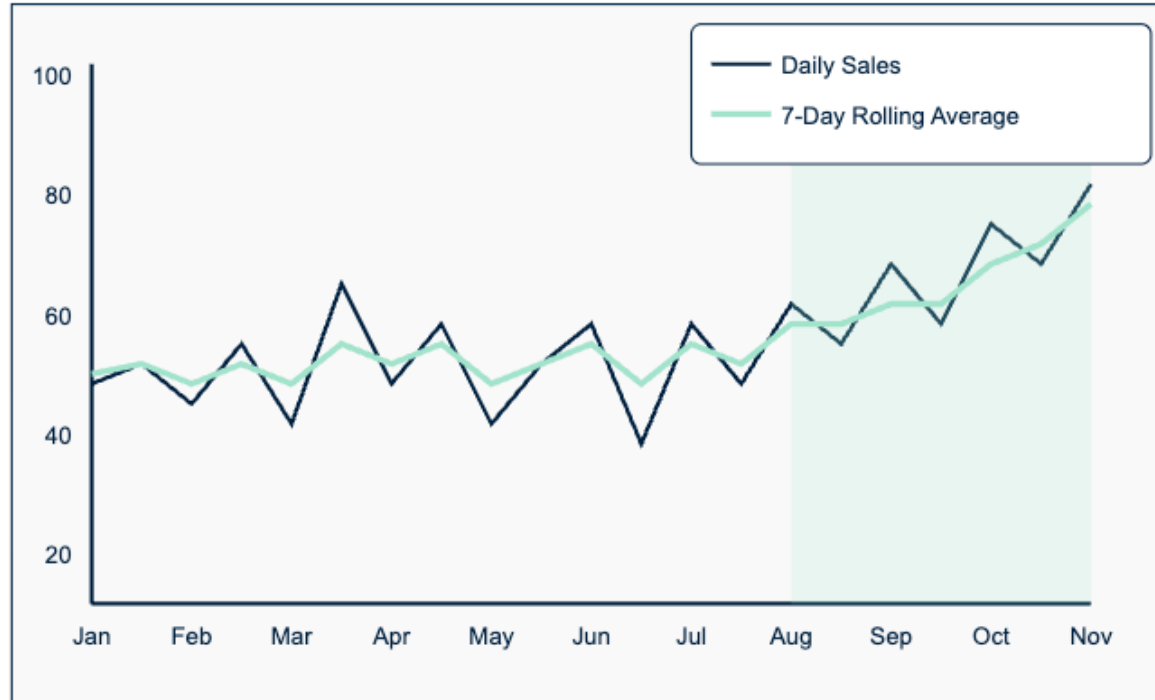
Descriptive Analytics Methods

From Data to Insights



A. Trend & Time-Based Analysis

Understanding how key metrics evolve over time



Key Techniques:

- Line charts (daily/weekly/monthly trends)
- Rolling averages (7-day, 30-day smoothing)
- Period-over-period comparison (WoW, MoM, YoY)
- Seasonality detection & time-series decomposition

Business Application:

An e-commerce company uses time-based analysis to identify:

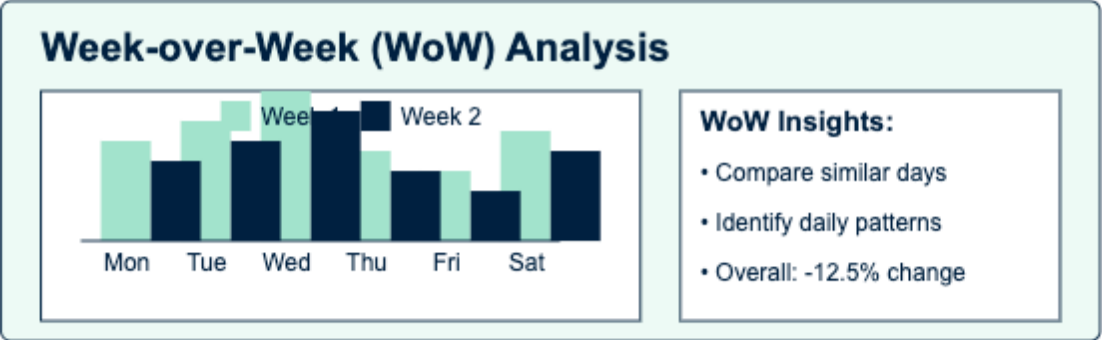
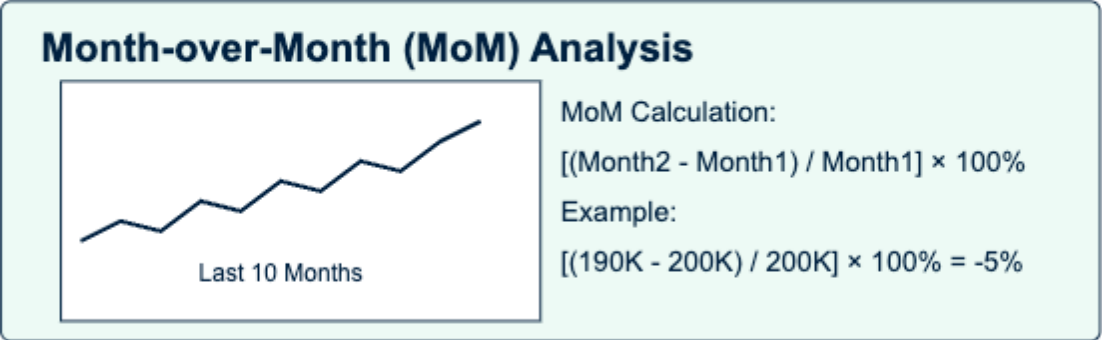
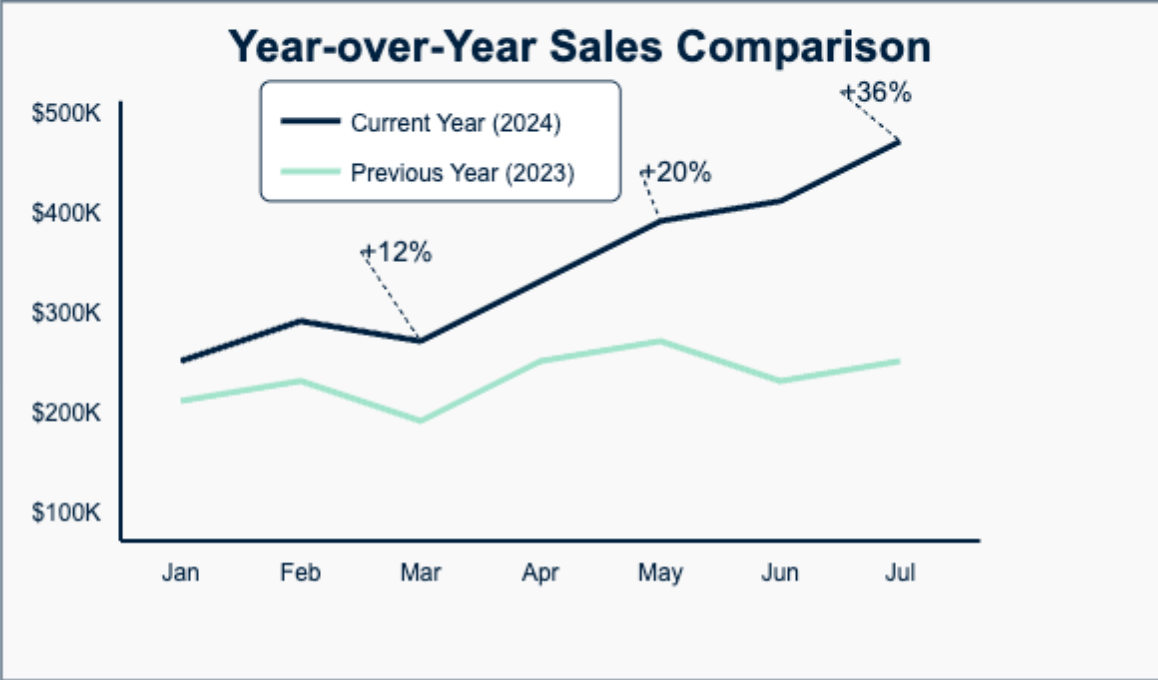
- Sales spikes during holiday seasons
- Day-of-week purchasing patterns
- Year-over-year growth rates

Implementation Tip:

Apply smoothing techniques (like 7-day rolling averages) to filter out noise and identify true patterns in volatile daily data.

Period-over-Period Comparison

Comparing performance across similar time intervals



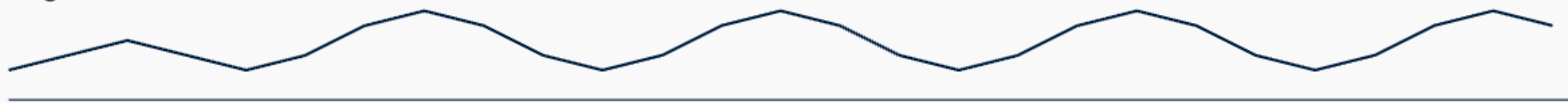
Business Benefits of Period-over-Period Analysis:

- Identifies growth trends and momentum (acceleration or deceleration)
- Reveals cyclical patterns and seasonality effects
- Measures marketing campaign effectiveness over similar time frames
- Normalizes for seasonal variations when comparing YoY same-period data

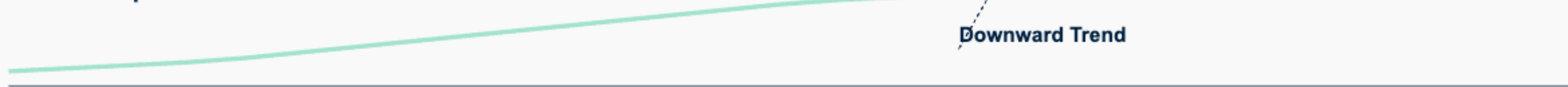
Seasonality Detection & Time-Series Decomposition

Breaking down data into trend, seasonal, and residual components

Original Time Series



Trend Component



Downward Trend

Seasonal Component



Weekly Pattern

Residual Component (Random Variation)



Additive Decomposition:

$$Y(t) = \text{Trend} + \text{Seasonality} + \text{Residual}$$

Business Applications:

- Accurate inventory planning for seasonal demand
- Marketing campaign timing optimization

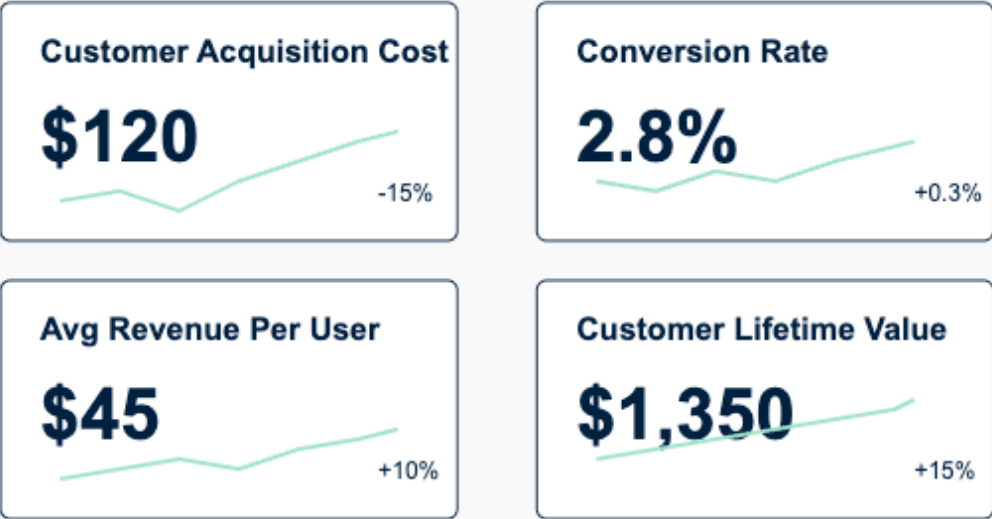
Implementation Notes:

- Common decomposition methods: Additive, Multiplicative
- Tools: Python (statsmodels), R (decompose), Excel (time series functions)

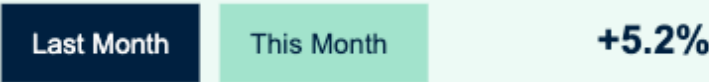
B. Summary Statistics & KPIs

Showing the current state of key business indicators

SaaS Executive Dashboard



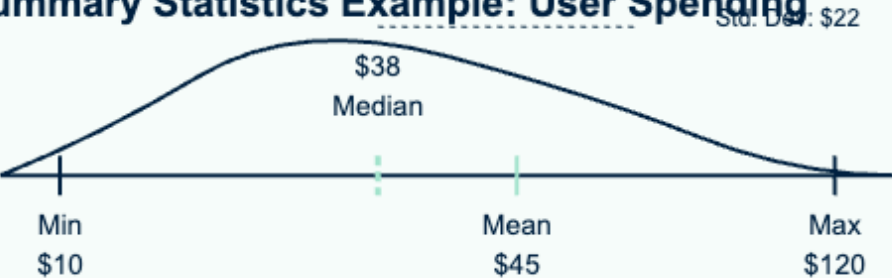
Monthly Growth Rate: +5.2%



Key Techniques:

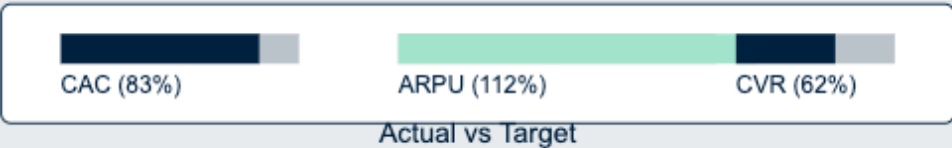
- Mean, median, min, max, standard deviation
- Growth rate (% increase or decrease)
- Cumulative totals
- Ratio metrics (CAC, CVR, AOV, ARPU)
- KPI scorecards & dashboards

Summary Statistics Example: User Spending



Implementation Tip:

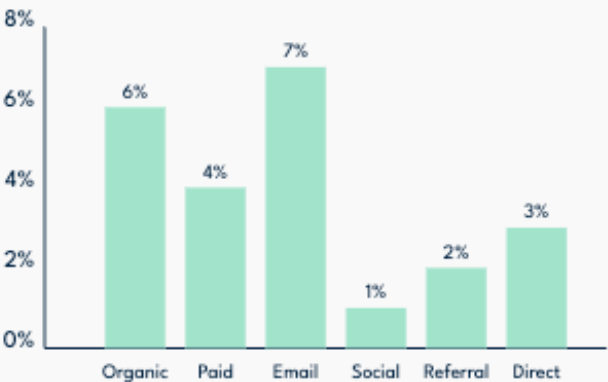
Design KPI dashboards that compare current metrics against targets and historical performance to provide context.



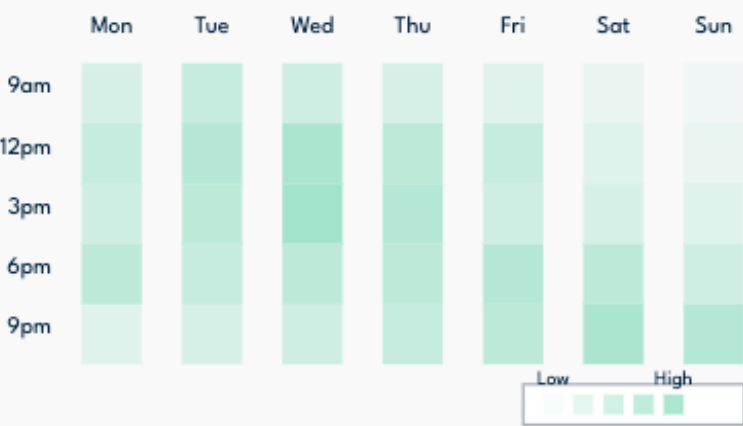
C. Segmentation & Grouped Analysis

Comparing how metrics differ across categories

Conversion Rates by Traffic Source



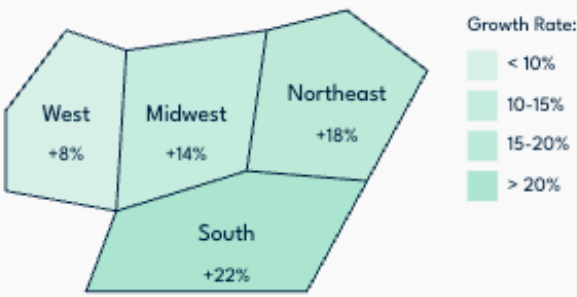
Click-Through Rate by Day & Hour



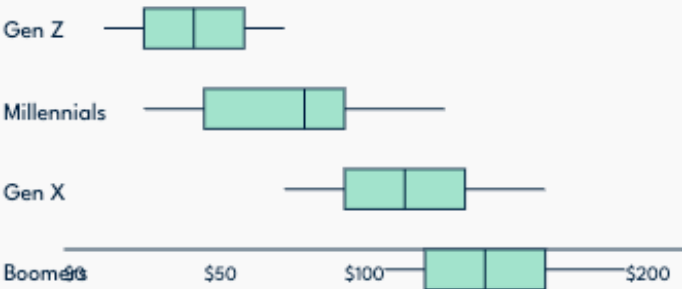
Key Techniques:

- Group-by analysis
- Bar charts (total/average per group)
- Pivot tables (cross-tabulated)
- Box plots (distributions)
- Heatmaps (multi-dimensional data)

Regional Performance Differences



Customer Spend by Demographic



Business Application:

- Conversion rates by traffic source
- ROI comparison across channels
- Customer spend by demographics
- Regional performance differences

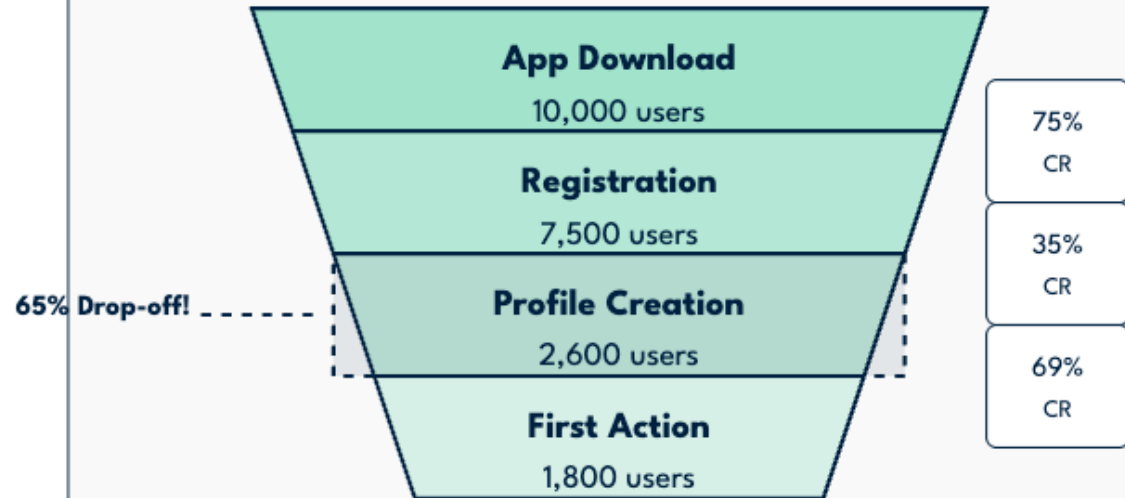
Implementation Tip:

Start with broad segments, then drill down into sub-segments to uncover more nuanced insights.

D. Funnel & Path Analysis

Measuring performance across step-based journeys

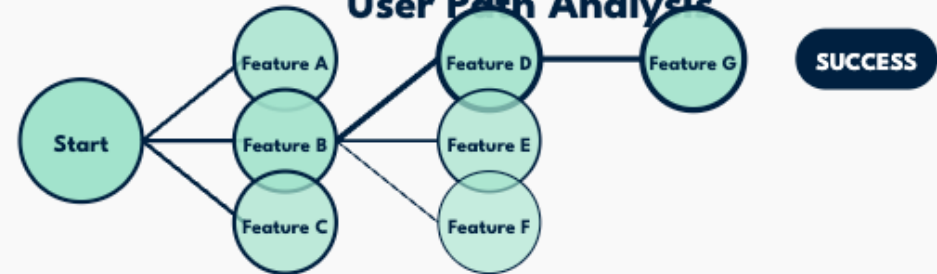
App User Onboarding Funnel



Key Techniques:

- Funnel charts (Impressions → Clicks → Signups → Purchases)
- Conversion rates at each step
- Drop-off rate analysis
- Path analysis (most common sequences of user behavior)

User Path Analysis



Most Successful User Path:

Start → Feature B → Feature D → Feature G

- 28% of all successful users follow this path
- 3.2x higher retention rate vs. other paths

Implementation Tip:

Focus optimization efforts on steps with the highest drop-off rates to maximize overall funnel performance.

Example: Simplify profile creation by reducing required fields to address the 65% drop-off at this stage.

E. Cohort & Retention Analysis

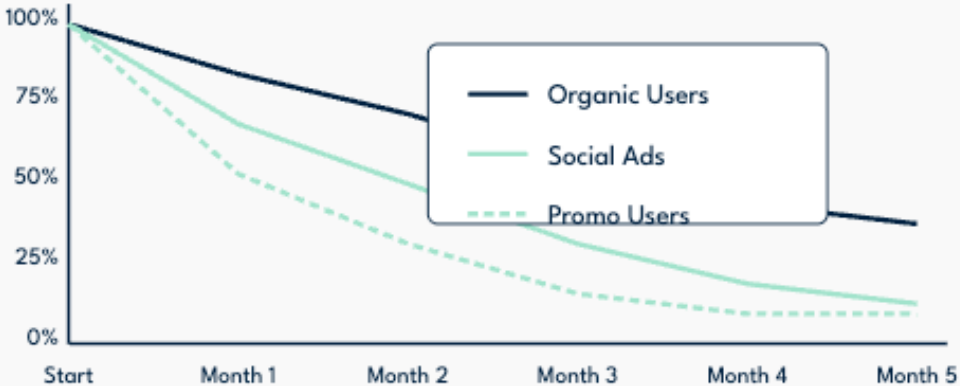
Tracking groups of users over time

User Retention by Signup Month

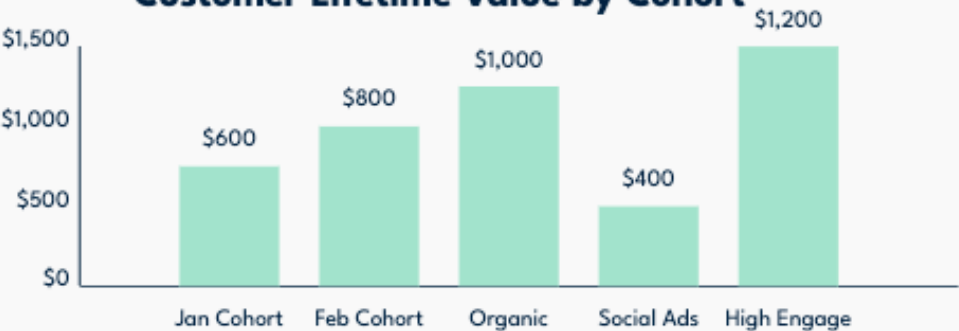
Cohort	M0	M1	M2	M3	M4	M5
Jan 2024	100%	82%	65%	48%	37%	32%
Feb 2024	100%	80%	62%	45%	33%	-
Promo Users	100%	68%	42%	30%	-	-
Organic Users	100%	84%	72%	56%	-	-

Retention % by Month Since Signup

Retention Curves by Acquisition Channel



Customer Lifetime Value by Cohort



Key Techniques:

- Cohort tables (user retention per signup period)
- Retention curves (active users over time)
- Cumulative retention/engagement
- CLTV by cohort

Business Application:

- Promotional vs organic retention
- Impact of product updates
- Highest-value acquisition channels
- Early engagement correlation with long-term retention

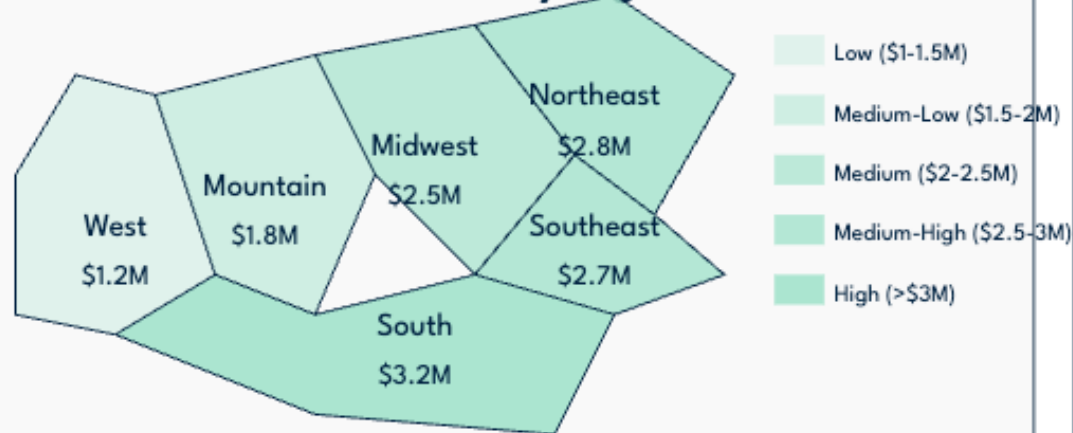
Implementation Tip:

Compare cohorts based on common attributes to identify factors that influence long-term value.

F. Geographic & Spatial Analysis

Understanding where performance varies by location

Sales Performance by Region



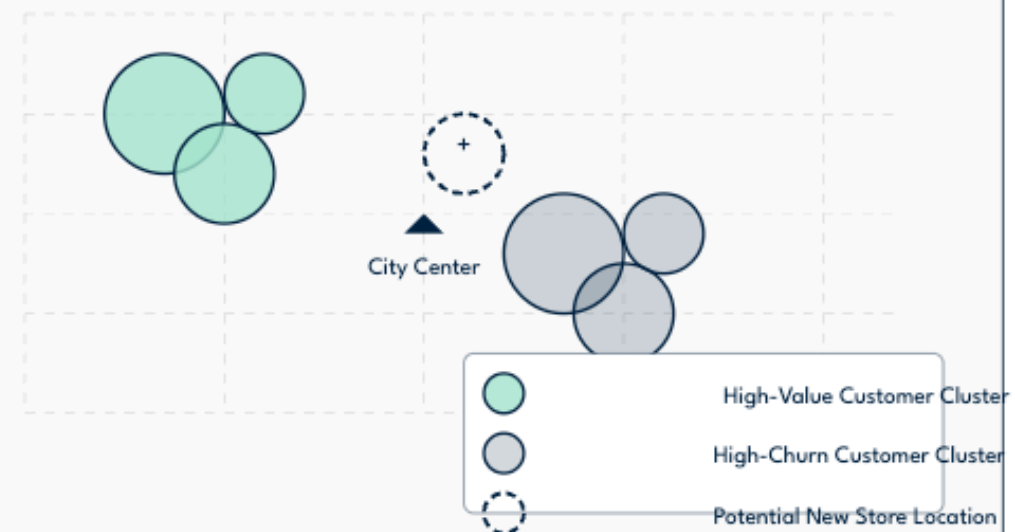
Average Order Value (AOV) by Region:

North: \$78

South: \$92

East: \$85

Customer Cluster Map (Urban Area)



Key Techniques:

- Geo heatmaps (sales by city or country)
- Regional comparisons (AOV in north vs. south)
- Cluster mapping (high-churn zones)

Business Application:

A retail chain uses geographic analysis to:

- Identify high-potential locations for new stores
- Optimize regional marketing spend
- Adjust inventory based on local preferences

Implementation Tip:

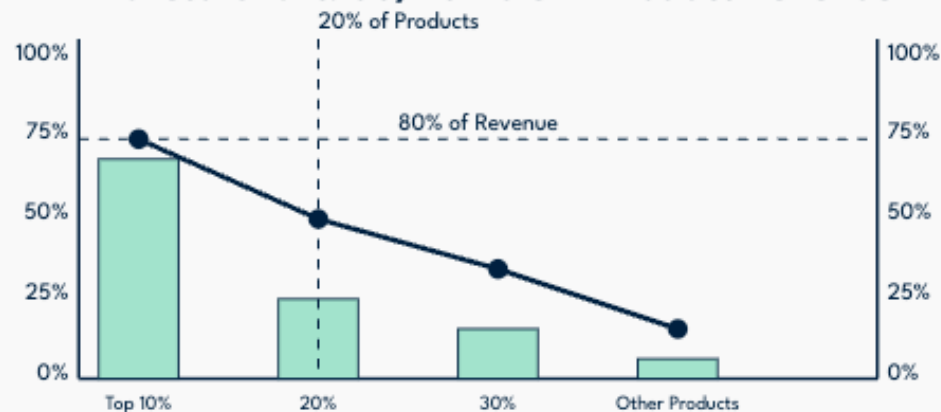
Combine geographic data with demographic information to understand the "why" behind regional performance differences.

Example: Higher-income areas → Higher AOV, Lower frequency

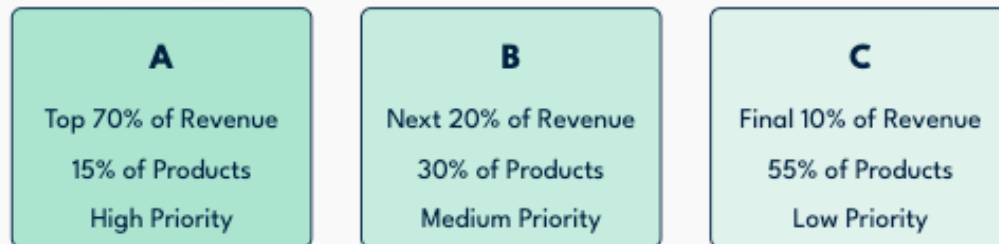
G. Product or SKU-Level Breakdown

Understanding which items drive performance

Pareto Chart: 80/20 Rule in Product Revenue

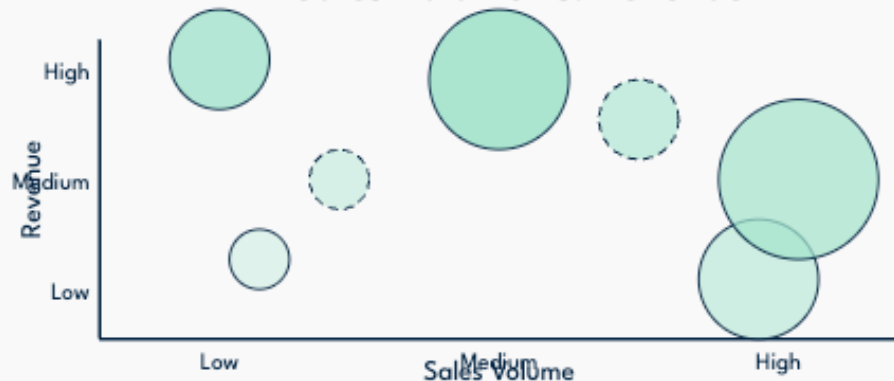


ABC Analysis: Product Segmentation



Category A: High inventory availability, premium shelf space, marketing focus
Category C: Consider discontinuing low performers, minimal stocking

Sales Volume vs. Revenue



Business Application:

A consumer goods company discovers:

- 15% of products generate 70% of profits
- High-volume items with low margins vs. low-volume with high margins
- Complementary product purchasing patterns
- Seasonal variation in product popularity

Key Techniques:

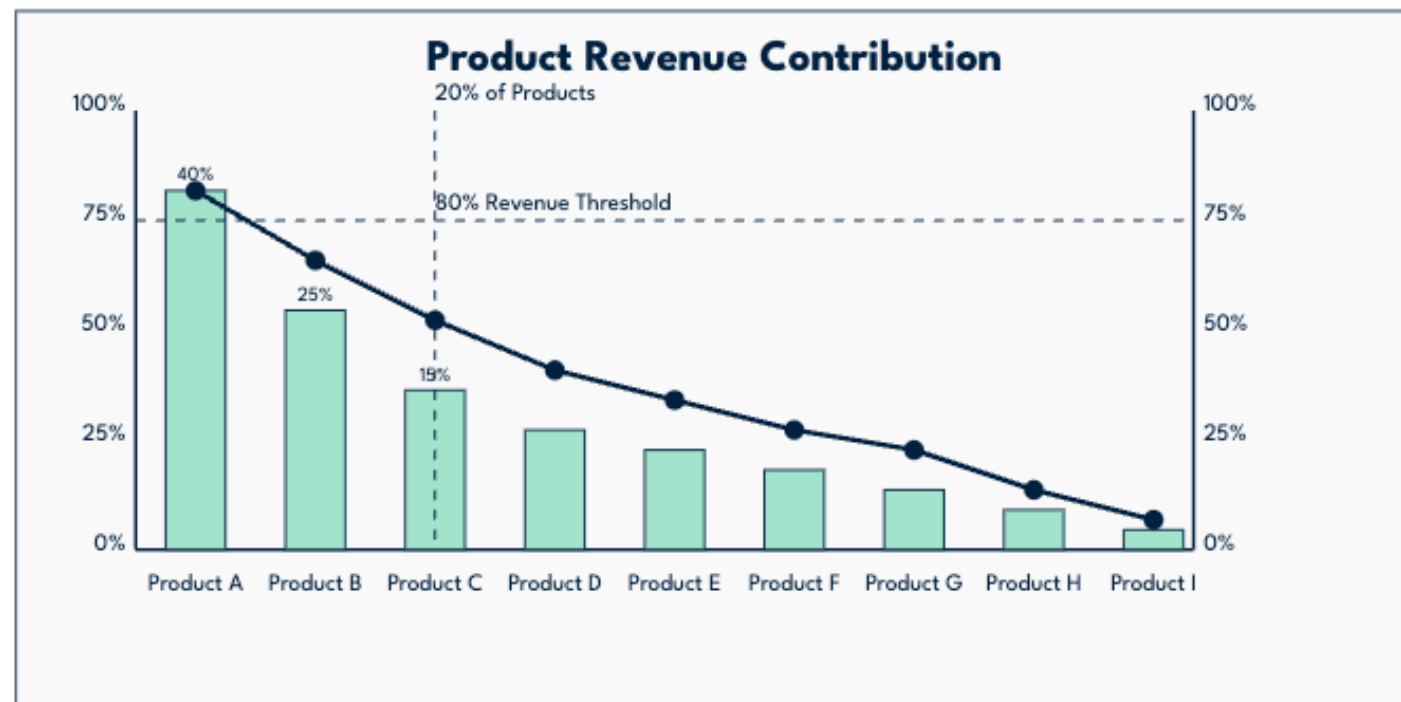
- Pareto charts (80/20 principle)
- ABC analysis (segment by revenue)
- Sales volume vs. revenue bubble charts
- Top/bottom performer tables

Implementation Tip:

Use ABC analysis to categorize products for inventory and marketing prioritization.

Pareto Chart

Visualizing the 80/20 Rule in Business Data



Understanding Pareto Charts

The 80/20 Principle:

- Roughly 80% of effects come from 20% of causes
- Named after economist Vilfredo Pareto
- Common pattern across many business domains

Key Components:

- Bar chart: Individual contribution (left Y-axis)
- Line chart: Cumulative percentage (right Y-axis)
- Items arranged from highest to lowest contribution
- Critical thresholds marked (typically 80% line)
- Focus on "vital few" vs "trivial many"

Business Applications

Product Management

- Identify top revenue-generating products
- Focus marketing on high-impact items
- Rationalize low-performing SKUs
- Prioritize inventory management

Quality Improvement

- Identify most common defect types
- Focus improvement on highest-impact problems first
- Track progress in reduction over time

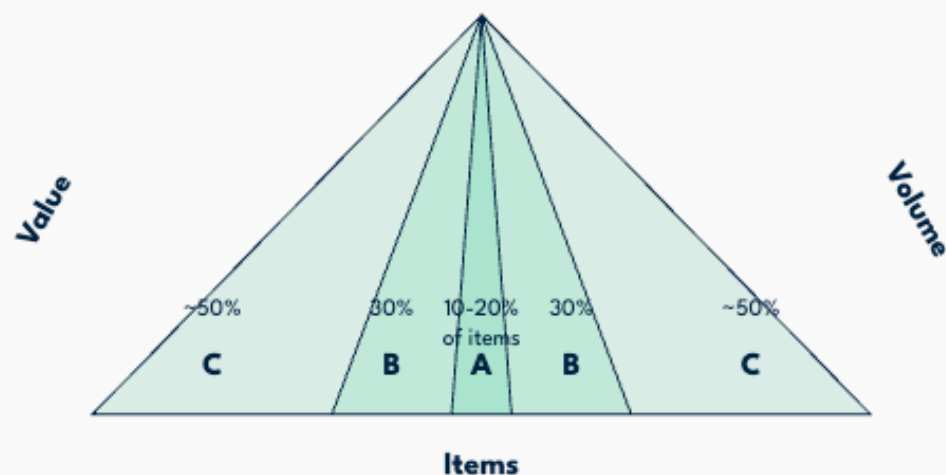
Customer Analysis

- Segment VIP customers (top 20%)
- Identify high-value customer behaviors
- Create tiered loyalty programs
- Develop targeted retention strategies

ABC Analysis

Segmenting Inventory by Value Contribution

ABC Inventory Classification



ABC Categories in Detail

A Items

Very Important
~70% of Value
10-20% of Items
Highest Control
Daily Monitoring
High Availability
Strategic Focus

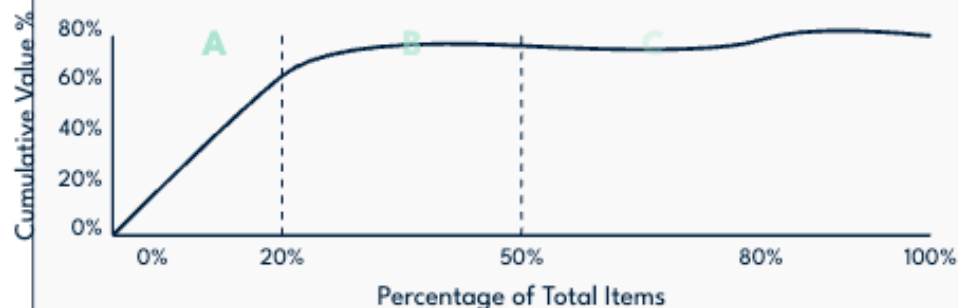
B Items

Moderately Important
~20% of Value
~30% of Items
Medium Control
Weekly Monitoring
Moderate Stock
Tactical Focus

C Items

Least Important
~10% of Value
~50% of Items
Minimal Control
Monthly Review
Limited Stock
Efficiency Focus

Value Distribution Curve



Implementation Best Practices

1. Data Collection & Analysis

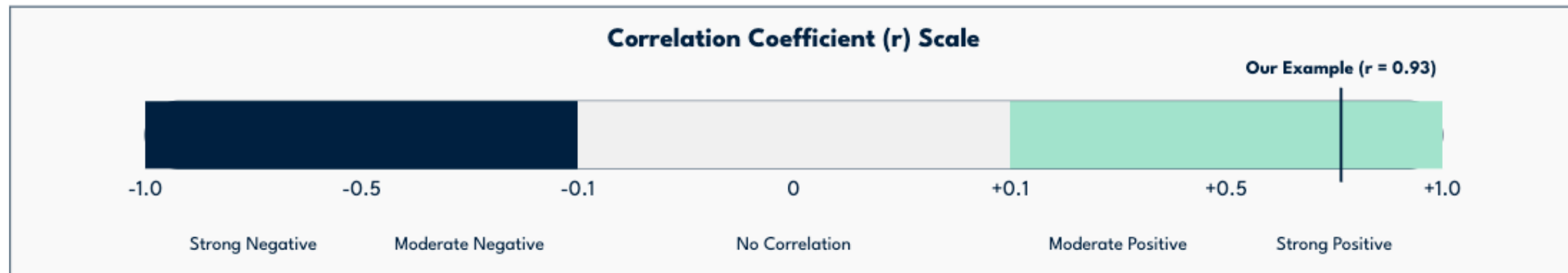
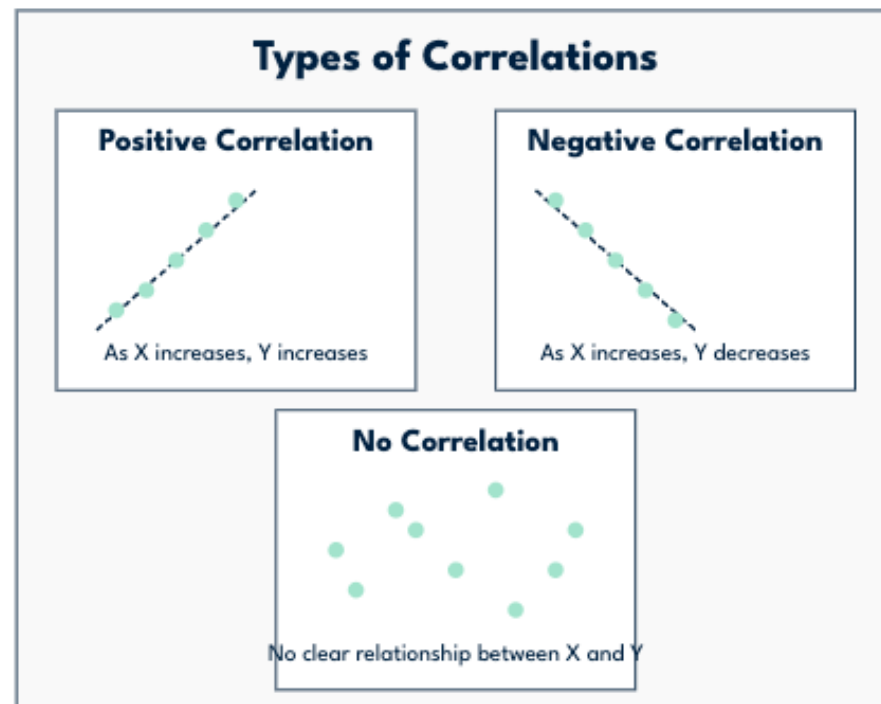
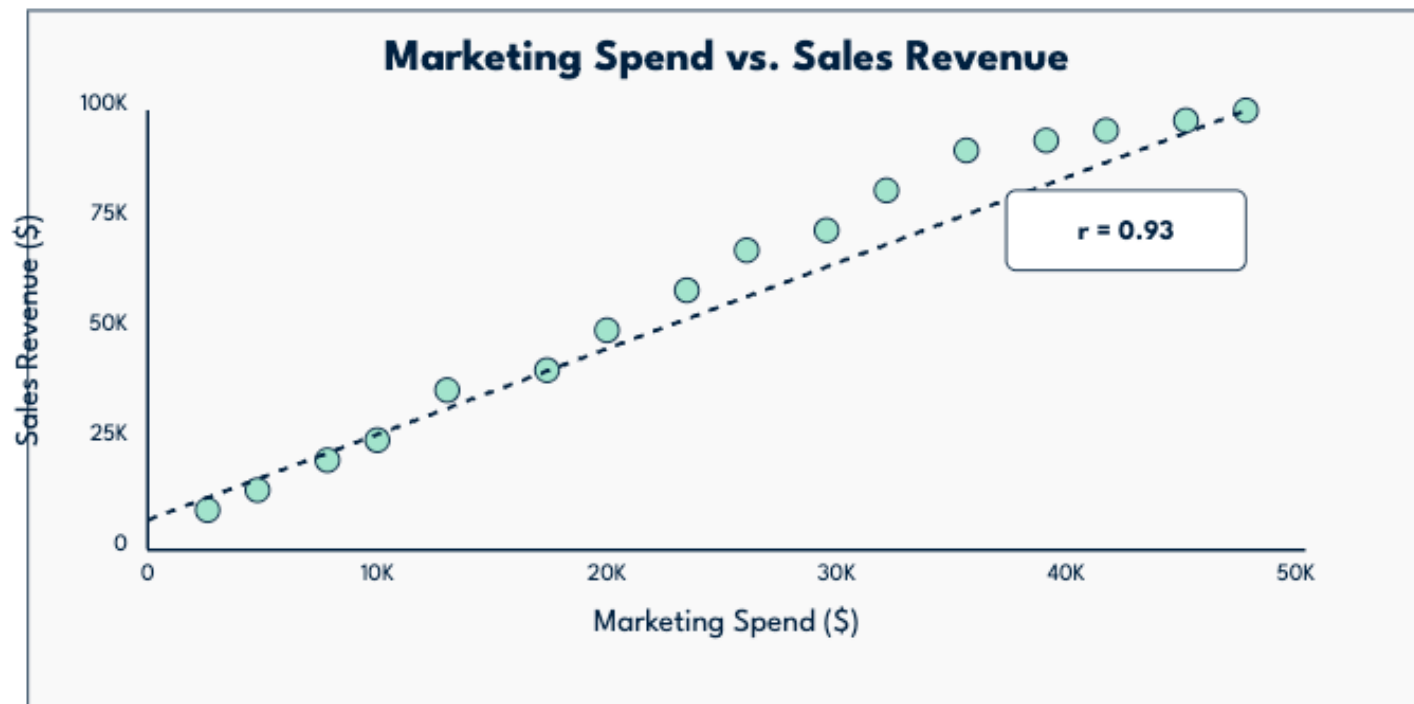
- Collect annual consumption value for each inventory item
- Arrange items in descending order of consumption value
- Calculate cumulative value and percentage of total items

2. Management Strategy by Category

- Customize inventory policies based on classification (A, B, or C)
- Focus resources on managing high-value, high-impact items

Correlation Analysis with Scatterplots

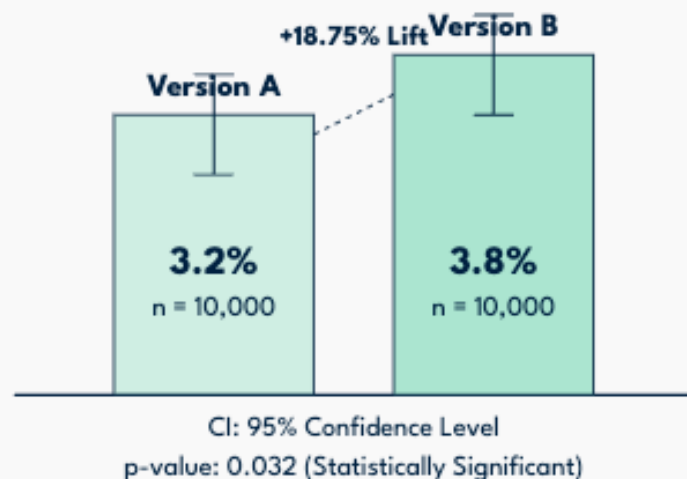
Visualizing Relationships Between Two Variables



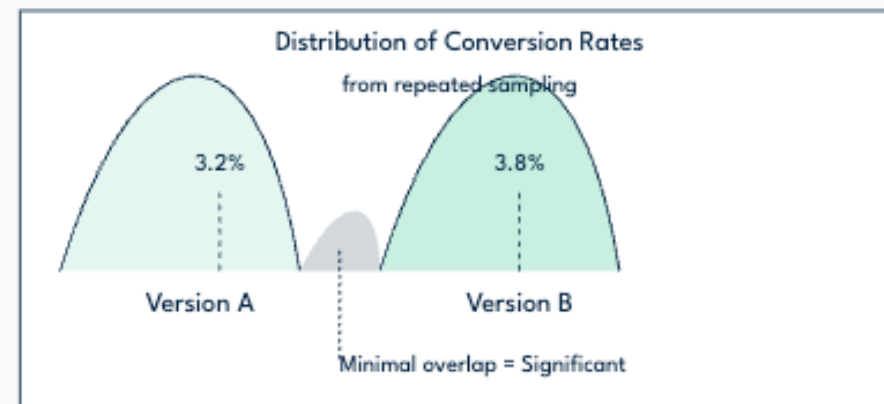
H. A/B Test Summaries

Seeing what happened in each test group

Product Page Conversion Rates



Understanding Statistical Significance



- p-value < 0.05 indicates less than 5% chance the result is due to random variation
- Minimal overlap between distributions suggests real difference between variants

Key Techniques

- Group-wise KPIs (conversion rate A vs. B)
- Lift calculation (% improvement)
- Difference charts with confidence intervals
- Statistical significance testing (p-values)

Implementation Tip

Balance Both Perspectives:

Statistical Significance

Is the difference real?

$$p < 0.05$$

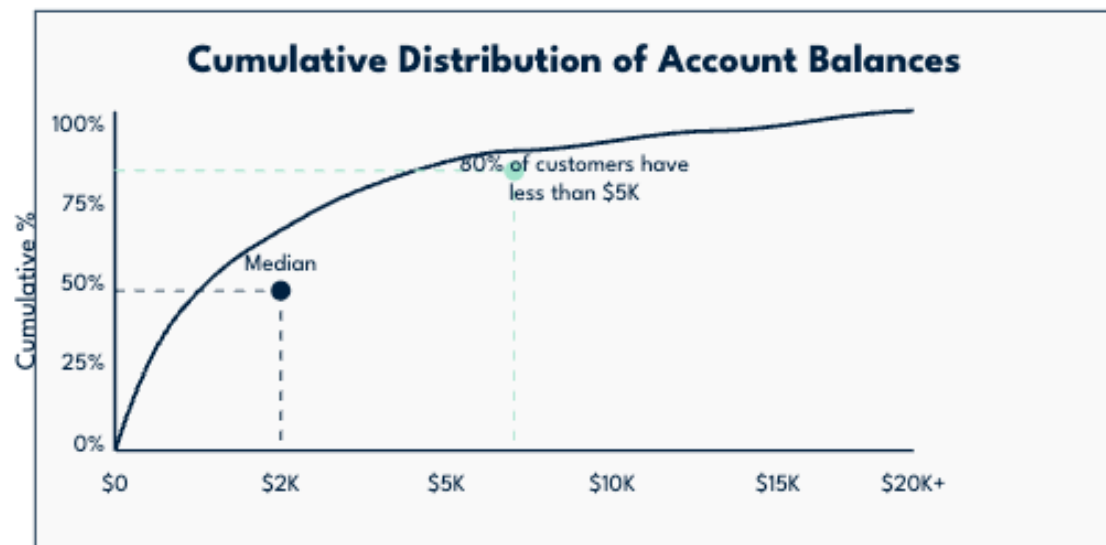
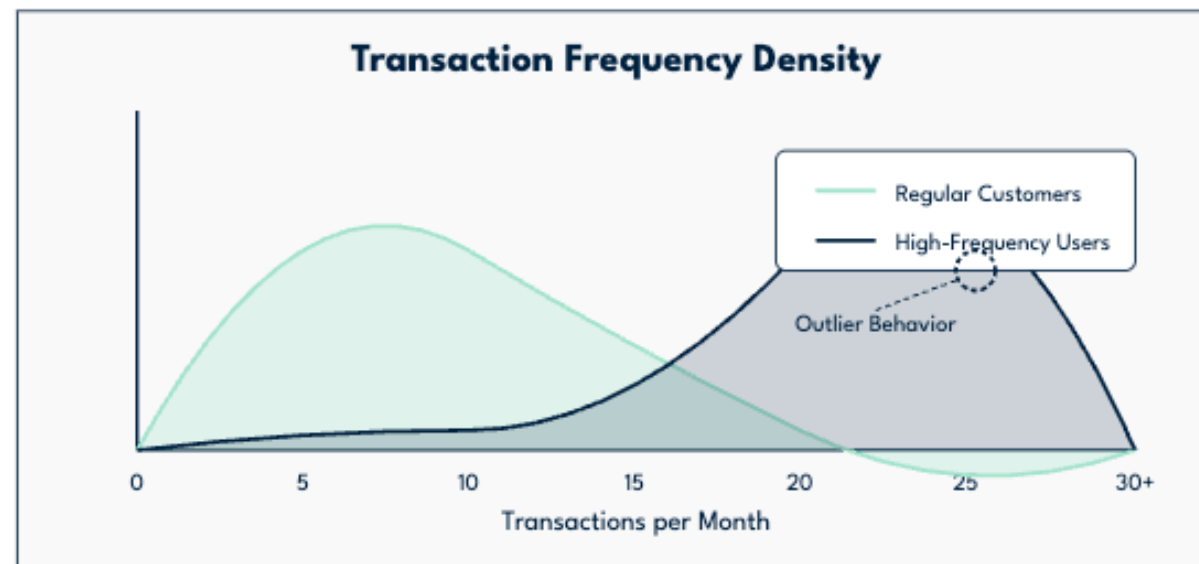
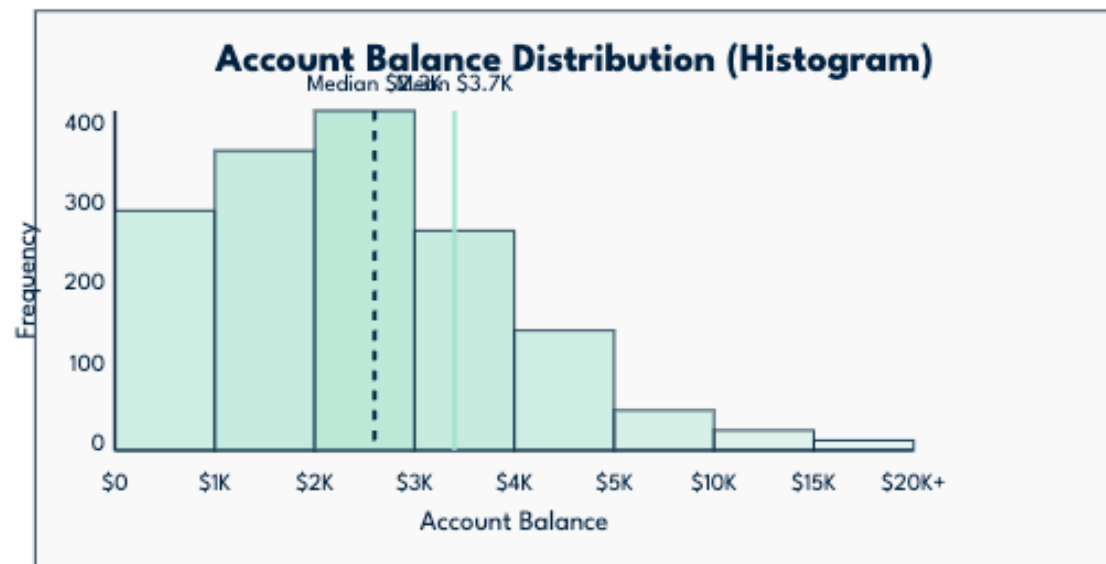
Business Significance

Is the difference meaningful?

ROI positive

I. Distribution & Frequency Analysis

Understanding how values are spread



Key Techniques

- Histograms (number of customers by order frequency)
- Density plots (smooth distributions)
- Frequency tables (counts per bin)
- Cumulative distribution curves
- Distribution statistics (mean, median, mode, variance)

Business Application

A financial services company analyzes:

- Account balance distribution
- Transaction frequency
- Outlier behavior identification
- Customer segmentation

Implementation Tip

Look beyond averages - outliers and distribution shape often reveal critical business insights.

J. Descriptive Text or Survey Summaries

Summarizing qualitative feedback

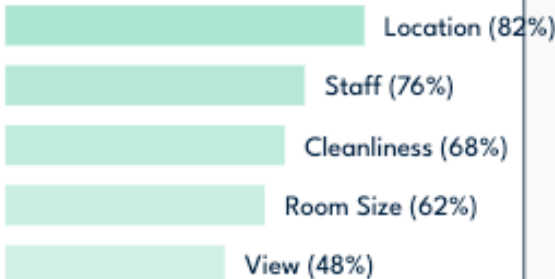
Guest Review Word Cloud



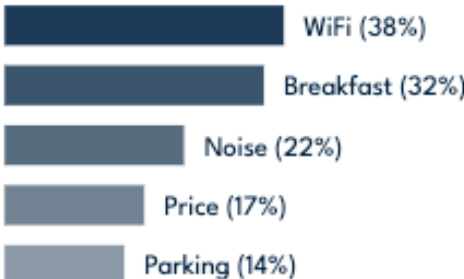
Positive Themes Negative Themes

Top Comment Themes Analysis

Positive Themes



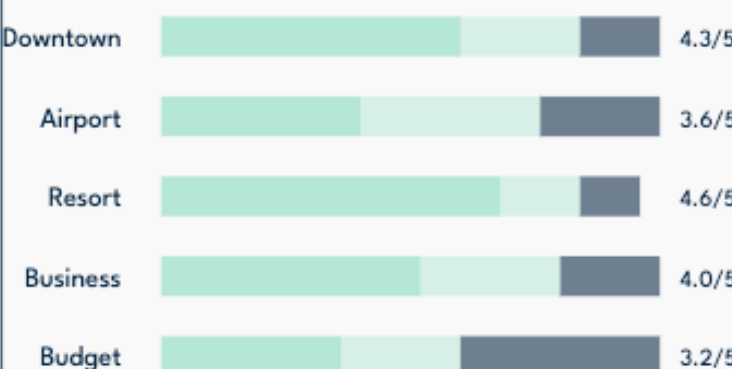
Negative Themes



Sentiment Trends Over Time



Feedback Comparison Across Properties



Key Techniques

- Word clouds (visual summaries)
- Theme analysis (keyword tagging)
- Sentiment scoring (pos/neg/neutral)

Positive Comments
Neutral Comments
Negative Comments

Bringing It All Together

An Integrated Approach to Descriptive Analytics

Descriptive Analytics Framework

1. Summary Statistics

Understand overall performance (KPIs, metrics)



2. Time-Based Analysis

Identify trends and patterns over time



3. Segmentation Analysis

Discover variations across different groups



4. Specialized Analysis

Apply specific techniques for deeper insights



5. Combine Quantitative + Qualitative Insights

Keys to Success



Begin with Clear Business Questions

Choose Appropriate Visualizations

Maintain Data Quality and Consistency

Provide Context for Interpretation

Connect Insights to Actionable Steps

Descriptive Analytics Toolkit

Trend & Time-Based Analysis

Summary Statistics & KPIs

Segmentation & Grouping

Cohort & Retention Analysis

Funnel & Path Analysis

Geographic & Spatial Analysis

Product/SKU Breakdown

Text & Survey Analysis